SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload for Admin Student Management

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload for Admin Student Management
Student Name	K.VARSHIKA
Student ID	JF/ICT/24/10
Implementation Status	Completed / Partially Completed / Pending — select the actual status
Date	2026/10/12

1. Change Request Summary

1.1 Business Requirement

The administrator could previously add and manage student information without uploading a profile image. The change is intended to let an authorized administrator upload a student profile image when creating a student record. The image should help identify students and should be displayed in the administrator's Students section and in the corresponding student's own profile after login.

1.2 Existing Behaviour

- The administrator can access the Student Management section.
- The administrator can add a student using the existing Add Student form.
- Student details are stored using the existing student record and student_id.
- The Add Student form did not provide a profile image upload field.
- Student records were displayed without profile images.
- Image file validation and image-path storage were not available.

1.3 Requested Change

The system must provide a profile image upload facility in the administrator's Add Student form.

- Allow PNG, JPG, and JPEG image files only.
- Reject unsupported files such as PDF, EXE, JS, PHP, HTML, and TXT, with a clear validation message.
- Validate the selected file in the frontend and validate it again on the backend.
- Store the uploaded image in a suitable server-side upload location and save its path/reference with the correct student record.
- Display each student's image beside the correct record in the administrator's Students section.
- After a student logs in using their registration number and password, display that student's own profile image.
- Show a default placeholder when no profile image is available.
- Preserve existing Add, Edit, Delete, login, and authorization behaviour.

2. Implementation

2.1 Implementation Summary

CR-010 introduces profile image upload and display for student records. The administrator selects an image while adding a student. The application validates the file type, stores the image, and associates the image path with the correct student record. The administrator's Students list displays the relevant image. The student dashboard retrieves and displays the profile image belonging to the logged-in student. The existing student_id or the project's established student-to-login relationship must be used to ensure that images are not mixed between students.

Complete this report only after checking the implementation in the running application. Replace any unverified status or result with the actual result.

2.2 Affected Files

File	Change	Purpose
frontend/src/pages/DashboardPage.jsx or existing Admin Students component	Modified if applicable	Add the image field, validate selected file, submit the image, and display images in the Students list.
frontend/src/styles/dashboard.css or relevant stylesheet	Modified if applicable	Style the upload field, profile image, placeholder, and validation message.
frontend/src/pages/StudentDashboardPage.jsx	Modified if applicable	Display the logged-in student's own profile image.
frontend/src/services/api.js	Modified if applicable	Send multipart/form-data and retrieve student profile data using the existing API setup.
backend/src/controllers/studentController.js	Modified if applicable	Process student creation/update and associate the saved image path with the correct student.
backend/src/routes/studentRoutes.js	Modified if applicable	Connect the student routes to the upload middleware and controller.
backend/src/middleware/uploadStudentPhoto.js	Added or modified if applicable	Validate and handle permitted image uploads.
backend/src/models/studentModel.js	Modified if applicable	Save and retrieve the profile image path/reference.
backend/server.js or backend/src/app.js	Modified if applicable	Register routes and configure static access to uploaded image files.
Database schema / students table	Modified if required	Add a profile image path/reference column if the table does not already have one.
backend/uploads/students/ or configured upload directory	Added/configured if applicable	Store uploaded student image files.

2.3 Technical Changes

Image Selection and Validation

- The upload control should accept PNG, JPG, and JPEG files.

- The frontend should check the selected file and show a clear message for unsupported formats.
- The backend must independently validate uploads; frontend validation alone is not sufficient.
- The backend should use safe server-generated filenames and apply a reasonable file-size limit.
- The uploaded file should be handled as an image and must not be executed as server-side code.

Image Storage and Database Association

- Store the image file in the configured upload directory or storage location.
- Store the image path/reference in the corresponding student record rather than storing the full file in a text field.
- Use the existing student_id or established student-account relationship to associate the image with the correct student.
- If no image is uploaded, retain a null/empty image reference and display the configured placeholder.

Administrator and Student Display

- Show the profile image in the Admin Students section beside the correct student.
- When a student logs in, obtain the profile belonging to that authenticated account; do not choose a student record based only on a client-supplied registration number.
- Provide a default image if the image path is missing or the file is unavailable.
- Ensure the image URL can be served by the application in both development and deployment environments.

2.4 Implementation Logic

1. The administrator opens the Add Student form.
2. The administrator enters the student's existing details and selects a profile image.
3. The frontend accepts PNG, JPG, or JPEG and displays an error for an unsupported file.
4. The frontend submits the student details and selected image to the existing backend route using the request format supported by the project.
5. The backend checks administrator authorization and validates the uploaded file again.
6. The backend saves the image with a safe filename and stores its path/reference with the correct student record.
7. The Admin Students section retrieves the student records and displays each available profile image or a placeholder.
8. The student logs in using the existing registration number and password.
9. The student dashboard retrieves the authenticated student's own profile and displays that student's image or a placeholder.
10. Existing Add, Edit, Delete, and login behaviour is tested to confirm it still works.

2.5 Error and Data Handling

- Reject unsupported formats and show a clear message such as: 'Only PNG, JPG and JPEG images are allowed.'
- Handle missing image selections without breaking student creation if images are optional in the approved requirements.
- Handle upload failures without saving an incorrect image path.
- Handle missing or unavailable image files by showing a placeholder.
- Avoid associating an image with the wrong student when adding or editing records.
- Keep existing student-loading and API error handling where possible.

3. Testing & Verification

3.1 Testing Approach

Testing should be performed manually in the development environment after the feature has been implemented. Record the actual results; do not mark a test as passed unless it has been performed.

Testing types: Functional testing, positive testing, file validation testing, UI testing, security/authorization testing, student-login testing, CRUD regression testing, and error handling.

Environment: Development
Browser: [Chrome / Edge / Firefox]
Test Roles: Administrator and Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open Add Student form	Profile Image field is displayed	[Record actual result]	Pass
TC-02	Upload a PNG image	PNG image is accepted and saved with the correct student	[Record actual result]	Pass
TC-03	Upload a JPG image	JPG image is accepted and saved with the correct student	[Record actual result]	Pass
TC-04	Upload a JPEG image	JPEG image is accepted and saved with the correct student	[Record actual result]	Pass
TC-05	Try uploading a PDF	File is rejected and a clear validation message is shown	[Record actual result]	Pass
TC-06	Try uploading EXE, TXT, JS, PHP, or HTML	Unsupported file is rejected by the server	[Record actual result]	Pass
TC-07	Add a student with a valid image	Student record and image reference are saved correctly	[Record actual result]	Pass
TC-08	View Admin Students list	The correct image is displayed beside the matching student	[Record actual result]	Pass
TC-09	Add a student without an image	Student creation follows the agreed optional-image behaviour and shows a placeholder	[Record actual result]	Pass
TC-10	Log in as a student	The logged-in student's own image is displayed	[Record actual result]	Pass
TC-11	Log in as a different	The second student's own	[Record actual	Pass

	student	image is displayed; images are not mixed	result]	
TC-12	Edit a student record	Existing student data and image association remain correct	[Record actual result]	Pass
TC-13	Delete a student record	Correct student is deleted and existing CRUD behaviour remains correct	[Record actual result]	Pass
TC-14	Attempt upload without administrator authorization	Unauthorized upload is denied	[Record actual result]	Pass
TC-15	Refresh/reopen the dashboard	Saved profile images still display correctly	[Record actual result]	Pass

3.3 Test Summary

Total Test Cases	15
Passed	15
Failed	0
Pass Rate	100%
Overall Test Result	pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Status
AC-001	The Admin Add Student form provides a profile image upload field.	Met
AC-002	PNG, JPG, and JPEG files are accepted.	Met
AC-003	Unsupported formats are rejected with a clear message.	Met
AC-004	File validation is performed on the backend as well as the frontend.	Met
AC-005	The uploaded image path/reference is saved with the correct student record.	Met
AC-006	The Admin Students section displays the correct student's image.	Met
AC-007	The logged-in student sees their own profile image.	Met
AC-008	A placeholder is displayed when no image is available.	Met
AC-009	Existing Add, Edit, and Delete functionality continues to work.	Met

AC-010	Existing administrator authorization and student login behaviour remain intact.	Met

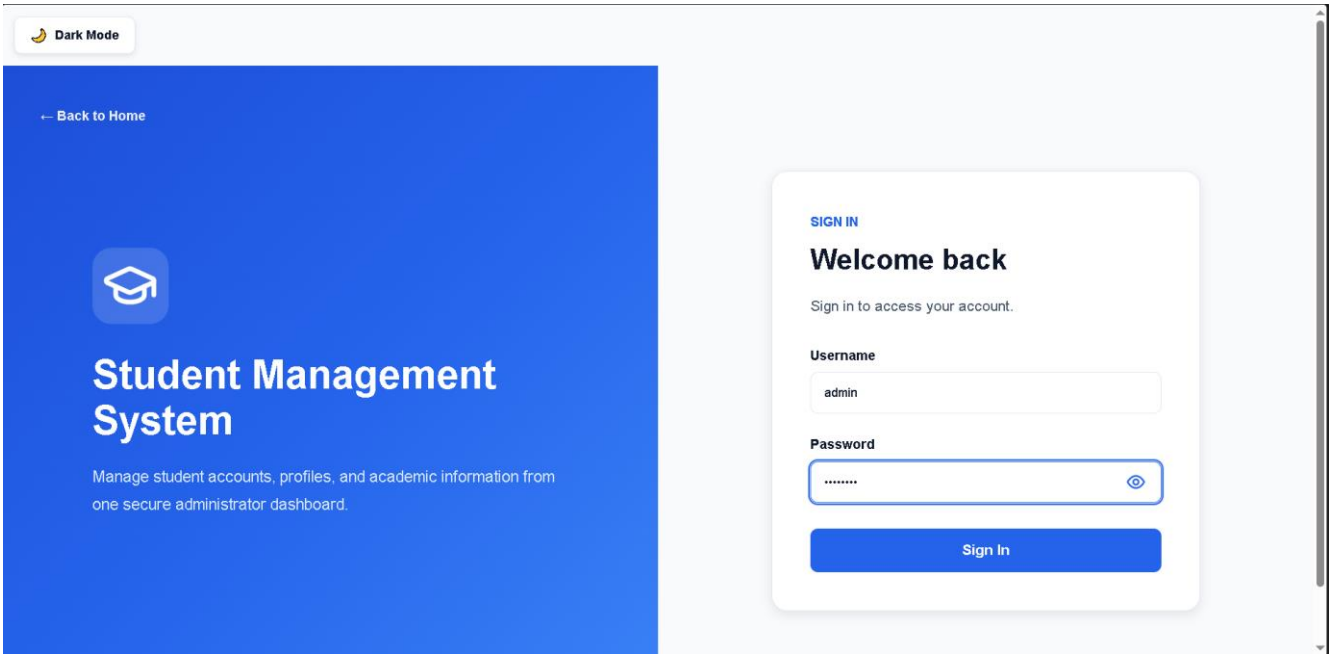
5. Evidence

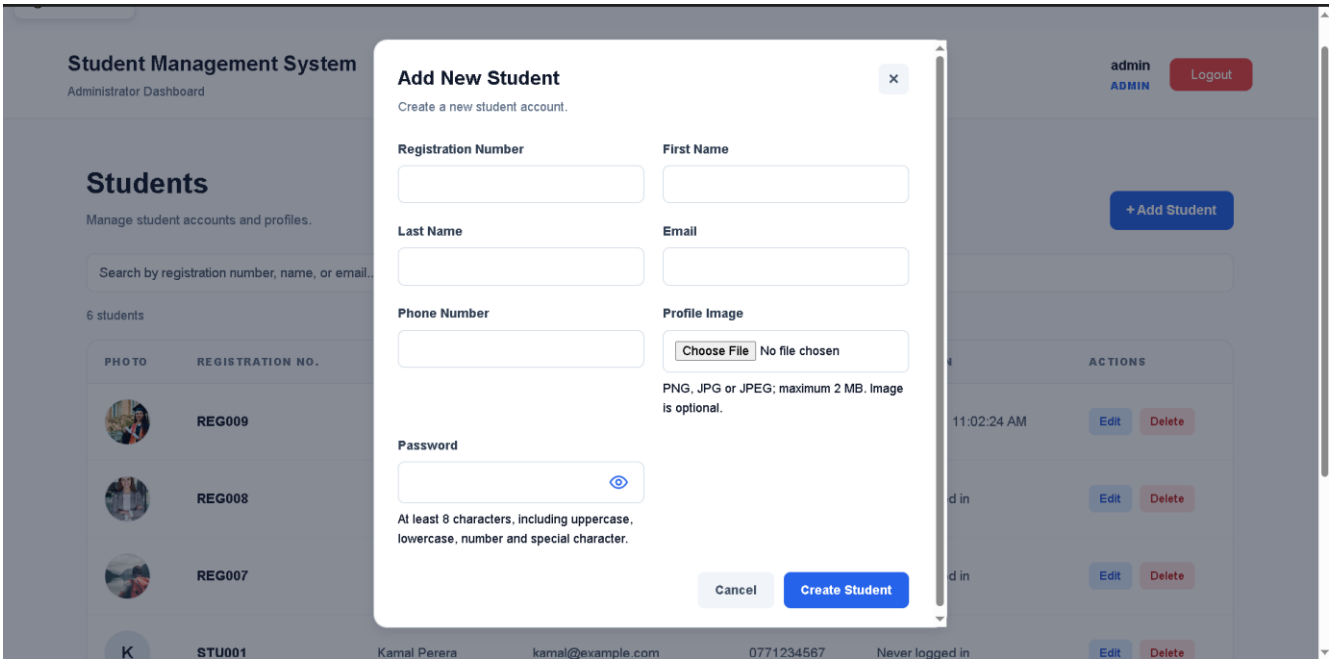
Attach up to two relevant screenshots as required by the report format. Use real screenshots from the implemented application.

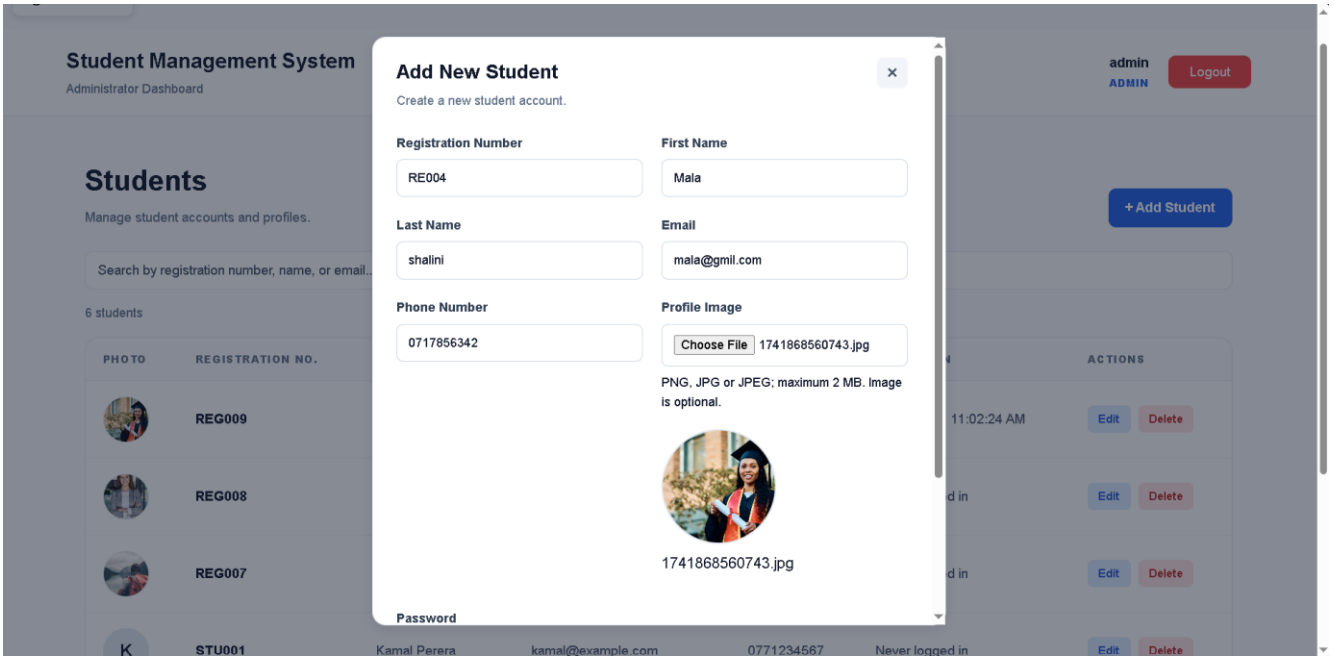
Evidence 01 – Admin Add Student and Students List

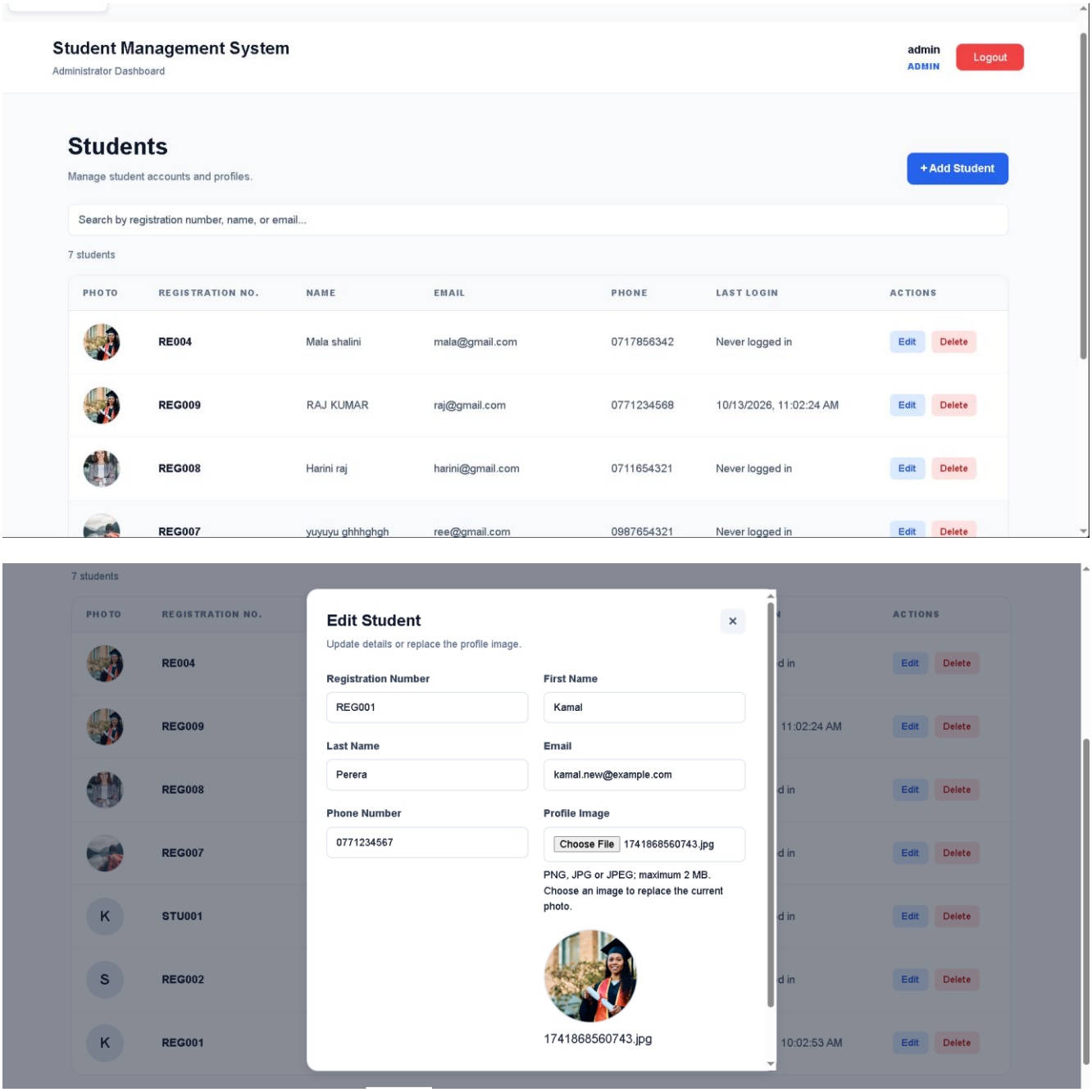
Show the Profile Image field, a supported image selected or uploaded, and the correct image displayed beside the corresponding student in the Admin Students section.

Figure 1: Admin Student Management showing profile image upload and the uploaded image beside the correct student.




Student Management System
Administrator Dashboard
admin
ADMIN
Logout
Students
Manage student accounts and profiles.
+ Add Student
Search by registration number, name, or email...
6 students
PHOTO
REGISTRATION NO.
ACTIONS
REG009
REG008
REG007
STU001
Kamal Perera
kamal@example.com
0771234567
Never logged in
Edit
Delete
Add New Student
Create a new student account.
Registration Number
First Name
Last Name
Email
Phone Number
Profile Image
Choose File
No file chosen
PNG, JPG or JPEG; maximum 2 MB. Image is optional.
Password
At least 8 characters, including uppercase, lowercase, number and special character.
Cancel
Create Student

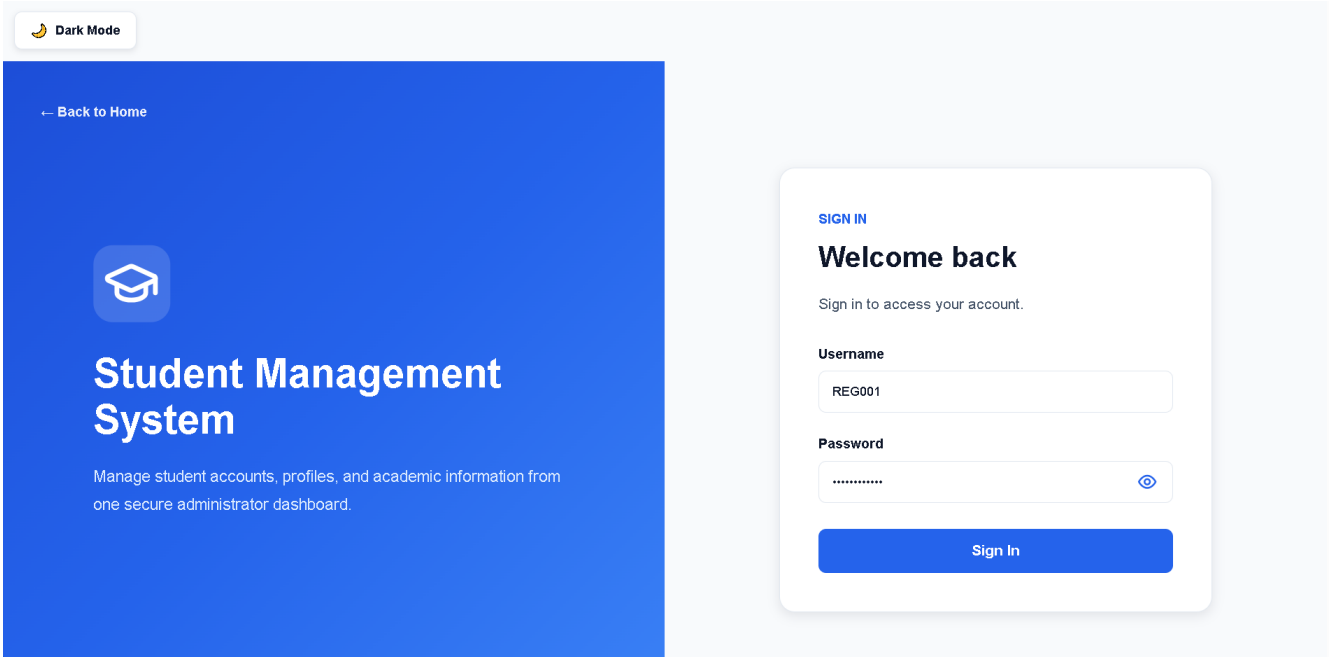

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Administrator Dashboard
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ADMIN
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+ Add Student
Search by registration number, name, or email...
6 students
PHOTO
REGISTRATION NO.
ACTIONS
REG009
REG008
REG007
STU001
Kamal Perera
kamal@example.com
0771234567
Never logged in
Edit
Delete
Add New Student
Create a new student account.
Registration Number
RE004
First Name
Mala
Last Name
shalini
Email
mala@gmil.com
Phone Number
0717856342
Profile Image
Choose File
1741868560743.jpg
PNG, JPG or JPEG; maximum 2 MB. Image is optional.
1741868560743.jpg
Password

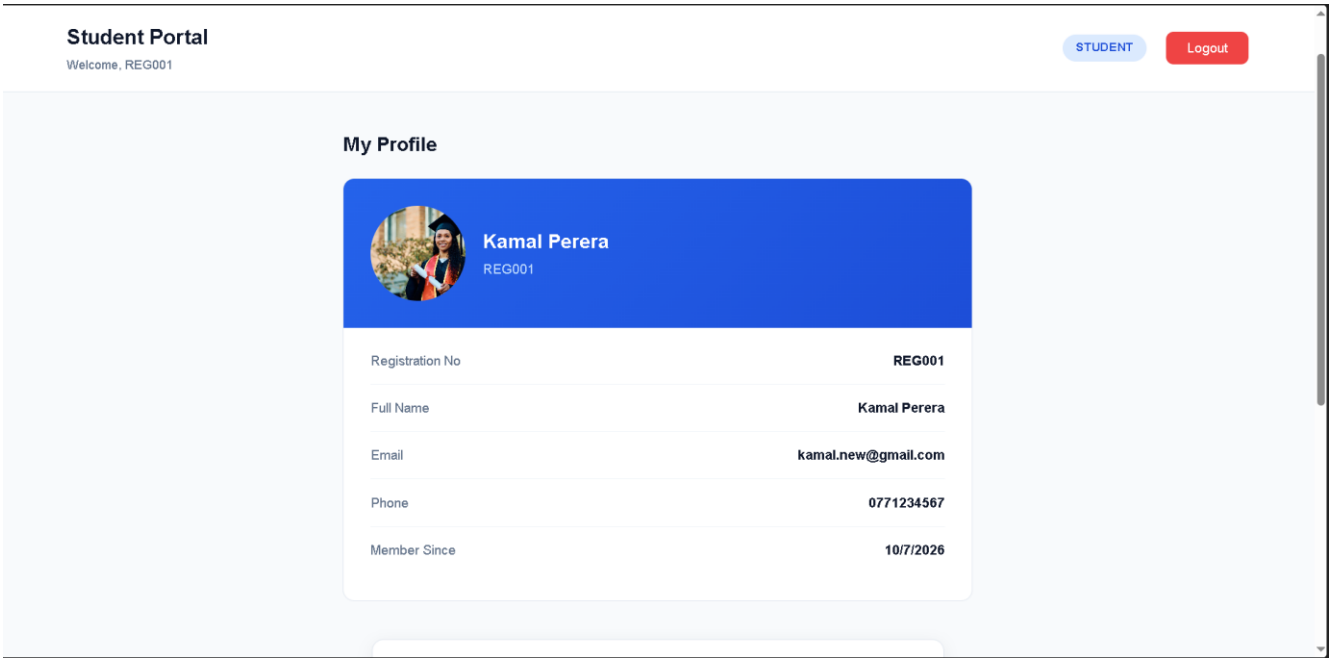


Evidence 02 – Validation and Student Dashboard

Show the validation message when an unsupported file is selected, and the logged-in student's own profile image on the student dashboard. If both cannot be shown clearly in one screenshot, use the most relevant evidence permitted by your assessment requirements.

Figure 2: Unsupported file validation and/or student dashboard displaying the profile image for the authenticated student.





6. Final Result

Requirements	[Met ~~/ Partially Met / Not Met~~]
Implementation	[Completed ~~/ Partially Completed / Pending~~]
Testing	[Passed / ~~Partially Passed / Failed / Not Yet Tested~~]
Acceptance Criteria	[All Met / ~~Partially Met / Not Met~~]
Existing CRUD	[Verified / ~~Not Verified~~]
Admin Students Image Display	[Verified / ~~Not Verified~~]
Student Login Image Display	[Verified ~~/ Not Verified~~]
Final Status	[Accepted ~~/ Pending / Rejected~~]

6.1 Final Summary

CR-010 adds profile image upload and display to the Student Management System. The administrator can select a student's profile image while adding a student. Only PNG, JPG, and JPEG formats should be accepted, and unsupported formats should be rejected with a clear validation message. The saved image path/reference must remain associated with the correct student record. The image should appear in the Admin Students section and in the profile of the student who logs in using their own account. A placeholder should be shown when no image is available. Existing student CRUD, administrator authorization, and student login behaviour must continue to work. The final implementation and test status must be confirmed using the actual application.

6.2 Issues / Limitations

[List any remaining issues or limitations. If none remain after testing, write: 'No significant issues identified during testing.']

Student Name: __K.VARSHIKA_____
Student ID: _____JF/ICT/_24/10_____
Date: _____12/10/2026_____
Signature: ___K.Varshika_____

~ END OF REPORT ~